

Assignment:- 4

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Branch: B.Tech CSE AI & ML

Semester: III

Subject: Machine Learning.

1) What is the time complexity of predicting with K-Nearest Neighbor?

→ The time complexity of predicting with KNN is $O(n \times d)$, where n is the number of training samples and d is the number of features. This is because, for each prediction, KNN calculates the distance between the test point and every training point. If sorting is required to find the top k nearest neighbours, it can go up to $O(n \times d + n \log k)$.

2) Explain the concept of precision in classification.

→ Precision measures how accurate the positive predictions of a classifier are. It's calculated as:

$$\text{Precision} = \frac{TP}{TP + FP}$$

where,

TP = True Positives

FP = False Positives

In simple terms, it tells us: "of all the instances predicted as positive, how many were actually positive?"
High precision means fewer false positives.

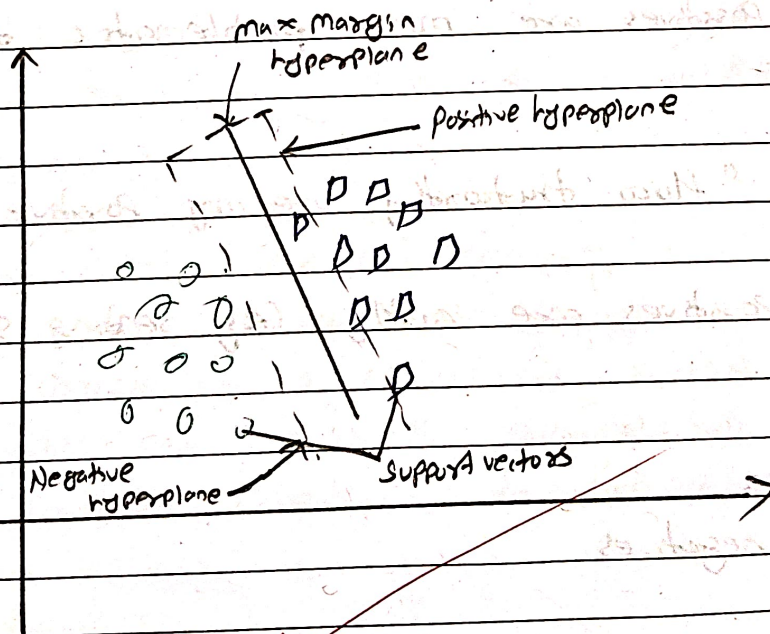
3) What is the space complexity of a linear regression model?

→ A linear regression model stores one weight for each feature and a single bias term. Therefore, its space complexity is $O(d)$, where d is the number of features. Once trained, it doesn't need to retain the training data, keeping memory usage minimal.

4) Define the Support Vector Machine (SVM) Algorithm.

→ Support vector machine (SVM) is a supervised learning algorithm that finds the optimal hyperplane separating data points of different classes while maximizing the margin between them.

— Support vector machine is used for classification and regression. It works by finding decision planes that separate data into different classes.



⑤ What is the time complexity of training a SVM?

→ Training a SVM involves solving a quadratic optimization problem, making it computationally heavy. The time complexity typically ranges between $O(n^2 \times d)$ and $O(n^3)$, depending on the dataset and kernel used. Linear SVM or approximate solvers can reduce this complexity for large datasets.

⑥ Explain the concept of Precision in classification.

→ Precision is the proportion of true positive predictions among all positive predictions made by the model.

$$\text{Precision} = \frac{TP}{TP + FP}$$

It focuses on correctness rather than completeness - useful when false positives are more problematic than false negative.

Interpretation: "How trustworthy are my positive predictions?"

Good for:

- when false positives are costly (e.g. sending spam mails to legit users).

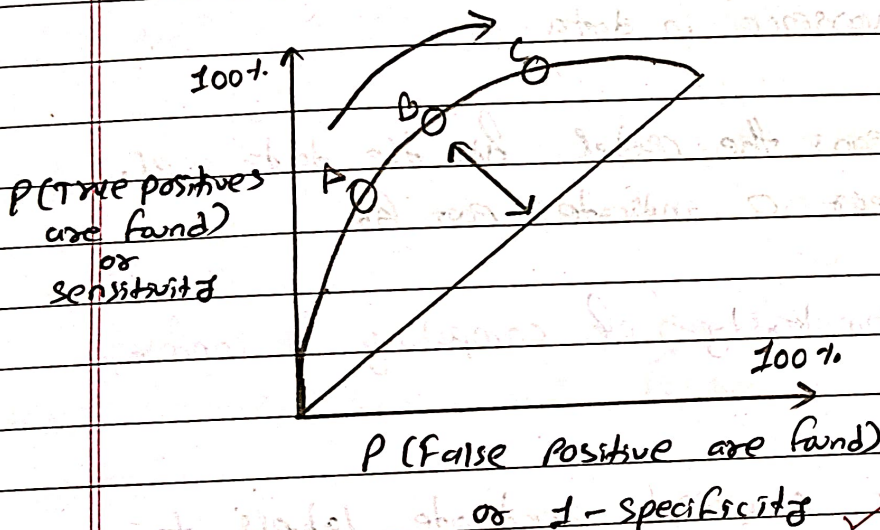
Weakness:

- Ignores false negatives

⑦ What is the purpose of the Roc curve in classification?

→ The Roc (Receiver operating characteristics) curve shows how the True Positive Rate (TPR) and False Positive Rate (FPR) change across different classification thresholds. It helps evaluate a model's ability to distinguish between classes.

The AUC (Area under curve) represents the model's overall performance - the closer to 1, the better.



⑧ What is the space complexity of a K-Nearest Neighbour model?

→ KNN is a lazy learner, meaning it stores the entire training dataset for use during prediction. Thus, its space complexity is $O(n \times d)$, where n is the number of samples and d is the number of features. Memory usage scales directly with dataset size.

⑨ Define the R^2 metric in regression

→ The R^2 (coefficient of determination) measures how well the regression model explains the variability of the target variable.

$$R^2 = 1 - \frac{SSE}{SST}$$

Where SSE = sum of squared errors

SST = total variance in data

An R^2 close to 1 means the model fits the data well, while ~~low~~ values near 0 indicate poor fit.

⑩ What is the time complexity of computing a confusion matrix?

→ The confusion matrix compares predicate labels to actual labels for n samples, so the time complexity is $O(n)$. It's a simple counting process that classifies each prediction into one of the four outcomes: TP, FP, TN, FN.

⑪ Explain the concept of recall in classification.

→ Recall (or sensitivity) measures how well a model identifies actual positives.

$$\text{Recall} = \frac{TP}{TP + FN}$$

It answers: "Of all true positives, how many did the model correctly detect?"

High recall means fewer false negatives - ideal for tasks like medical diagnosis.

(12) What is the space complexity of a support vector machine model?

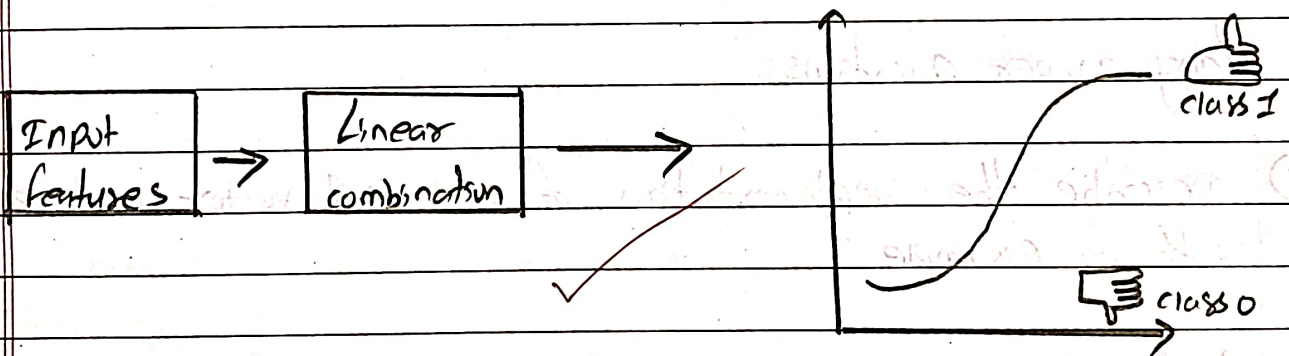
→ The space complexity of mainly depends on the number of support vectors after training. It's approximately $O(s \times d)$, where s is the number of support vectors and d is the number of features. For non-linear kernels, storing the kernel matrix can rise to $O(n^2)$.

(13) Define a classification task with an example.

→ A classification task assigns data samples into predefined categories based on their features.

Example:

Predicting whether an email is spam or not spam using features like word frequency, sender and subject line.



(14) What is the time complexity of evaluating a regression model?

→ Evaluating a regression model (like computing RMSE, MAE or R^2) requires comparing predicted and actual values for all samples.

Hence,

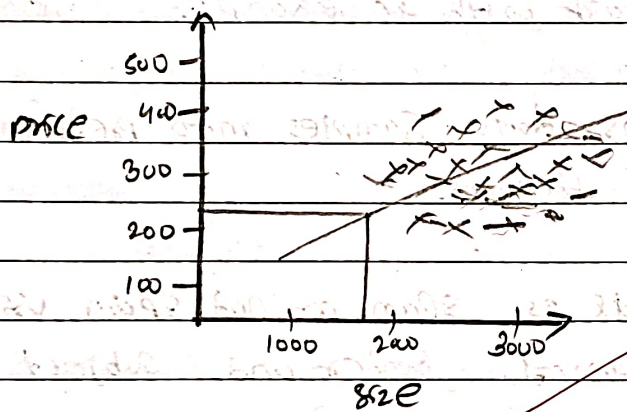
the time complexity is $O(n)$, where n is the number of test samples.

⑤ Explain the concept of a regression task with an example

→ A regression task predicts a continuous numerical value rather than a class label.

Example: Predicting house prices based on features like size, location, and ~~human~~ number of rooms.

The goal is to minimize the difference between predicted and actual values.



Long answer questions:

1) Describe the implementation of a support vector machine with an example.

→ A SVM is a supervised machine learning algorithm primarily used for classification and regression tasks. It works by finding an optimal hyperplane that best separates data points of different classes with the maximum margin (distance between the hyperplane and the nearest data points).

Implementation steps:

1) Data Preparation:

Load and clean the dataset, handle missing values, and normalize features since SVMs are sensitive to feature scaling.

2) Splitting the data:

Divide the dataset into training and testing sets.

3) Model Training:

Use an SVM classifier (like `SVC` from `sklearn`) and choose a kernel - eg. Linear, polynomial, or RBF depending on data patterns.

4) Prediction & Evaluation:

Example:

```
from sklearn import datasets
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.svm import SVC
```

```
from sklearn.metrics import accuracy_score
```

```
X, y = datasets.load_iris(return_X_y=True)
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

```
model = SVC(kernel='linear')
```

```
model.fit(X_train, y_train)
```

```
predictions = model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, predictions))
```

Like classifying emails as spam or not spam based on textual features.

② Explain the role of the ROC curve in evaluating classification models.

→ The ROC (Receiver operating characteristics) curve is a graphical representation of a classification model's performance across different threshold settings. It plots:

- True Positives Rate (TPR) on the Y-axis
- False Positives Rate (FPR) on the X-axis

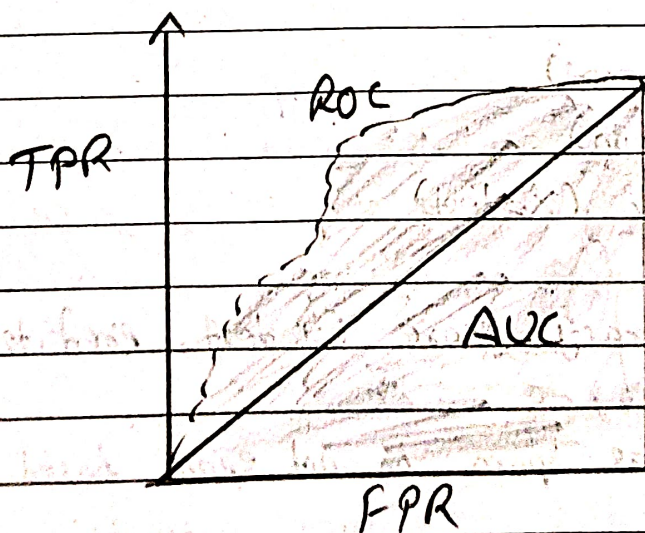
Each point on the curve represents a different threshold for deciding whether a sample belongs to the positive class.

Purpose:

- Shows the trade-off between sensitivity (recall) and specificity
- Helps compare multiple models.
- A model that performs well has a curve closer to top-left corner.

AUC (Area under curve):

- The AUC value summarizes the overall performance
- $AUC = 1 \rightarrow$ perfect classifier
- $AUC = 0.5 \rightarrow$ Random guess
- Higher AUC means better model performance



③ Discuss the time and space complexity of k-Nearest Neighbour (KNN).

→ Time complexity:

- Training: $O(1)$ → KNN is a lazy learner; it doesn't train explicitly.
- Prediction: $O(nd)$ → Each test sample requires distance calculation with all n training samples having d features.
Sorting neighbours adds $O(n \log k)$ overhead.

Space complexity:

- $O(nd)$ since KNN stores the entire training dataset for predictions.

Trade off: High memory and slower predictions, but easy to implement and highly interpretable.

④ Describe the implementation of grid search for hyperparameter tuning.

→ Grid search is a brute-force method used to find the best hyperparameter combination for a model.

Steps:

1) Define the model:

Choose the algorithm (e.g. SVM, Random Forest)

2) Create a parameter grid:

List the possible values for each hyperparameter

3) Cross-validation

use k-fold cv to evaluate model performance for every parameter combination.

4) Select Best Parameters:

The combination with the highest cross-validation accuracy is chosen.

Example:

```
from sklearn.model_selection import GridSearchCV
```

```
from sklearn.svm import SVC
```

```
Param_grid = {'C': [0.1, 1, 10], 'kernel': ['linear', 'rbf']}
```

```
grid = GridSearchCV(SVC(), Param_grid, cv=5)
```

```
grid.fit(X_train, y_train)
```

```
Print ("Best parameters:", grid.best_params_)
```

③ Explain the role of MME metric in regression evaluation

⇒ MME refers to mean magnitude of error or mean model error like similar to MAE / MSE

The MME (mean model error) is used to measure how far the predicted value are from the actual ones in regression tasks. It captures the average prediction error magnitude

$$MME = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- y_i = actual value, $\hat{y}_i$ = predicted value

Interpretation:

- Lower MME means better model performance
- It is simple, interpretable, and less sensitive to outliers than MSE

⑥ Explain unsupervised learning. Discuss clustering and dimensionality reduction techniques.

→ unsupervised learning deals with unlabeled data, the algorithm identifies hidden patterns or structure without explicit supervision.

i) clustering:

Grouping similar data points together based on feature similarity

- k-means clustering: Divides data into k clusters by minimizing intra-cluster variance.

- Hierarchical clustering: Builds nested clusters in a tree structure

- DBSCAN: Detects clusters based on density, great for irregular shapes.

ii) Dimensionality Reduction:

Reducing the no of input features while retaining key info.

- PCA (Principal Component Analysis): Projects data into fewer orthogonal dimensions.

- t-SNE / UMAP: used for visualization of high-dimensional data

- Autoencoders: Neural Network that compresses and reconstruct data

④ Explain the concept of Reinforcement Learning with a suitable real-life example.

→ Reinforcement learning is a type of learning where an agent interacts with an environment, learns through trial and error, and gets rewards or penalties based on its actions.

Key components

- Agent: learner or decision-maker
- Environment: everything the agent interact with
- Action: what the agent can do
- State: The current situation of the environment
- Reward: feedback signal from the environment

Example: self-driving cars learn to navigate roads by getting positive rewards for staying in the lane and reaching destinations safely, and negative rewards for collisions.

⑤ Describe any three machine learning applications in business and industry.

→ 1) Fraud detection (Finance):

ML models like Random forests detect abnormal transaction patterns.

2) Customer segmentation (Marketing):

- Clustering group customers based on purchase behavior for promotions.

3) Predictive maintenance (Manufacturing):

- Regression and time-series models predict equipment failures, reducing downtime and maintenance cost.

⑨ Discuss the importance of data preprocessing in the machine learning process.

→ Data preprocessing transforms raw data into a clean and usable format for ML models.

It's crucial because poor-quality data = poor model performance.

Key Steps:

- Data Cleaning: Handle missing values, outliers, and inconsistencies.
- Data Transformation: Normalize or standardize features.
- Encoding: Convert categorical variables into numerical form.
- Feature Selection: Remove redundant or irrelevant features.

Importance:

- Improves accuracy and model stability.
- Reduces training time.
- Enhances generalization to unseen data.

Without preprocessing, even the best algorithm can fail.

⑩ Explain the advantages, disadvantages and limitations of machine learning?

Advantages

- Automation: Learns patterns and automates complex decision-making.
- Continuous Improvement: Improves with more data and retraining.
- Broad Applicability: Used across healthcare, finance, education etc.
- Handle complex data: Can manage high-dimensional and non-linear data.

Disadvantages:

- Data Dependency: Needs large, quality datasets to perform well.
- High computation cost: Training advanced models requires significant resources.
- Interpretability: Some models (like deep learning) acts as "black boxes".

Limitations:

- Bias & ethics: models can inherit bias from data.
- Overfitting: models can perform well on training data but fails on new data.
- Security risks: vulnerable to data poisoning and adversarial attacks.

[Signature]